Machine Learning (CS-433) – Project 1

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***Abstract*—We predict self-reported myocardial infarction or coronary heart disease (MICHD) from BRFSS 2015 survey responses. Our pipeline combines survey-aware cleaning, missingness indicators and logistic regression trained with mini-batch stochastic gradient descent. Stratified five-fold cross-validation yields F1 0.430 ± 0.004 and ROC AUC 0.861; the final AIcrowd submission scores 0.444. Threshold selection and broader feature coverage provide the largest observed gains. Interactions and alternative recodings improve F1 only slightly, motivating the simpler final model.**

I. Introduction

We predict whether a respondent has previously been diagnosed with myocardial infarction or coronary heart disease (MICHD). The provided BRFSS 2015 subset [1] contains 328,135 labelled observations with 321 features and 109,379 unlabelled test observations. The target records past diagnoses, rather than future disease onset. Positive-class F1 is the primary evaluation metric; we also report the area under the ROC curve (AUC).

II. Models and methods

A. Exploratory data analysis

MICHD-positive respondents represent 8.83% of the labelled observations. Missingness is extensive: 99 features are missing in more than 90% of observations, 121 in more than 70%, and 147 in more than 50%. Mean missingness is 44.12% for positive and 44.85% for negative respondents. These averages do not identify a missingness mechanism. Conditional questions and optional survey modules create structural missingness, motivating missingness indicators. Features include continuous, discrete and categorical variables.

Prevalence increases with age and differs by sex, hypertension and diabetes status (Fig. 1). These descriptive patterns motivate testing interactions; they do not establish causal effects.

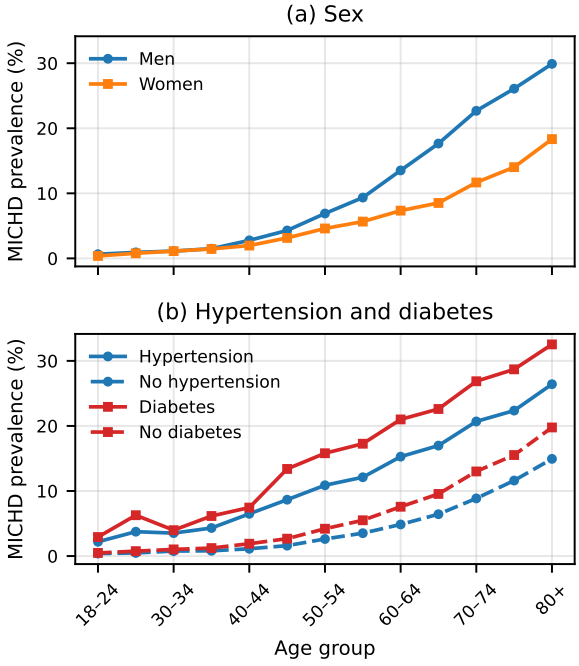


Figure 1. Unweighted MICHD prevalence in the training data by age group, stratified by (a) sex and (b) hypertension and diabetes status.

B. Preprocessing

The pipeline follows the types, relevance groups and survey-code rules in `variables_BRFSS_2015.csv`.

Feature selection: We compare cumulative groups 1 through N, with $N \in \{1, \ldots, 5\}$: *established risk factors* (Group 1, 10 variables: age, sex, BMI, smoking, alcohol, physical activity, hypertension, high cholesterol, diabetes and chronic kidney disease [2], [3], [4]); *documented associations* (Group 2, 22 variables: comorbidities, treatments, general health and socioeconomic status); *additional detail* (Group 3, 47); *optional modules* (Group 4, 64); and *exploratory variables* (Group 5, 60, e.g. vaccinations and seat-belt use). These groups encode prior judgments about relevance. We exclude target-revealing variables (`CVDINFR4`, `CVDCRHD4`, `HAREHAB1`), administrative fields and, by default, 74 alternative recodings.

Cleaning and decoding: Missing-answer codes become NaN; genuine-zero codes become zero (e.g. 88 in `PHYSHLTH` means zero days). Variable-specific recodings are applied. Only `DROCDY3_` and `_DRNKWEK`, stored in hundredths, are rescaled by 0.01; `_BMI5` is already in BMI units. Structured frequencies and durations are converted to one unit per variable; month–year dates become month indices. Special responses, such as "never smoked regularly", receive binary indicators and missing numerical entries.

Encoding and standardization: Binary categories become one $0/1$ column; nominal variables are one-hot encoded without their most frequent reference category. Categories observed fewer than 50 times are pooled. Ordinal

and numerical variables retain their values and are median-imputed. We add a binary missingness indicator for each feature with missing values, discarding entirely missing features and constants without missing values. Every encoded column, including binary indicators, is centered and divided by its standard deviation, using a scale of one when the deviation is zero.

C. Training

Logistic regression predicts

$$p(\mathbf{x}) = \frac{1}{1 + \exp[-(\mathbf{w}^T\mathbf{x} + b)]}, \tag{1}$$

with coefficients $\mathbf{w}$ and intercept b. Labels become $\{0, 1\}$ during optimization. We minimize mean binary cross-entropy plus $\lambda\|\mathbf{w}\|_2^2$, without penalizing b; the final model uses $\lambda = 0$.

Mini-batch stochastic gradient descent starts from zero parameters and shuffles observations for each of 30 epochs, with batch size 256 and learning rate $\eta_t = 0.05/\sqrt{t+1}$, starting at $t = 0$. An initial optimizer comparison found log-loss differences only at the fourth decimal place, motivating these fixed settings. The threshold maximizes F1 on training predictions (approximately 0.208 after fitting all labelled data); final predictions use the original $\{-1, 1\}$ labels.

D. Validation and model comparisons

We use stratified five-fold cross-validation and seed 38 for both folds and training. All learned preprocessing statistics, category levels, rare-category pooling, imputation, interaction choices, model parameters and thresholds use only the training portion of each fold. F1 and AUC are evaluated on held-out observations.

The `cv_experiments.py` script compares cumulative groups, $\lambda \in \{0, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$, a fixed threshold of 0.5, and alternative recodings, excluding unsupported raw formats. It also evaluates five products of encoded Group 1 features, with and without squared age. Pairs are ranked by the absolute logistic-loss gradient at the fitted linear model; added terms use training-set standardization.

Baselines use the final features: an always-negative classifier; ridge regression with $\lambda = 10^{-4}$, labels ± 1 and an F1-tuned score threshold; full-batch logistic regression (300 updates, learning rate 0.5); and a linear SVM. The SVM uses $\lambda = 10^{-4}$ and positive-class weight $n_-/n_+ \approx 10.3$ (negative weight one), with 30 epochs, batch size 256 and rate $0.01/\sqrt{t+1}$. Its threshold also maximizes training F1. The final pipeline refits groups 1–5 on all labelled data with $\lambda = 0$, without nonlinear terms or alternative recodings, producing 691 encoded columns.

III. Results

Table I reports F1 mean and standard deviation across the five folds, and mean AUC. Adding Group 2 provides the largest feature gain: F1 rises from 0.366 to 0.420 (ΔF1 $= +0.054$). Groups 3–5 add another 0.010 overall. Threshold selection raises F1 from 0.241 to 0.430 ($+0.189$), with unchanged AUC because the underlying scores are identical.

Table I
STRATIFIED FIVE-FOLD CV: F1 MEAN ± STANDARD DEVIATION AND MEAN AUC. UNLESS STATED OTHERWISE, THRESHOLDS MAXIMIZE TRAINING F1. FINAL: GROUPS 1–5, LOGISTIC SGD AND $\lambda = 0$.

Experiment	F1	AUC
Group 1	0.366 ± 0.005	0.820
Groups ≤ 2	0.420 ± 0.004	0.856
Groups ≤ 3	0.427 ± 0.004	0.859
Groups ≤ 4	0.428 ± 0.003	0.860
Groups ≤ 5 (final)	$\mathbf{0.430 \pm 0.004}$	**0.861**
Final, threshold 0.5	0.241 ± 0.003	0.861
Final, $\lambda = 10^{-5}$	0.430 ± 0.004	0.861
Final, $\lambda = 10^{-4}$	0.430 ± 0.004	0.861
Final, $\lambda = 10^{-3}$	0.430 ± 0.004	0.861
Final, $\lambda = 10^{-2}$	0.425 ± 0.005	0.859
Final + 5 interactions	0.432 ± 0.003	0.861
Final + interactions + age^2	0.432 ± 0.003	0.861
Final + alternatives	0.432 ± 0.003	0.861
Always negative	0.000 ± 0.000	0.500
Ridge regression	0.419 ± 0.003	0.854
Logistic, full-batch GD	0.430 ± 0.004	0.861
Linear SVM (class weights)	0.427 ± 0.004	0.860

AIcrowd F1, final submission: 0.444.

Penalties up to 10^{-3} give essentially unchanged F1; $\lambda = 10^{-2}$ lowers it to 0.425. Five interactions give paired $\Delta\text{F1} = 0.0016 \pm 0.0018$; adding squared age gives 0.0017 ± 0.0017, each improving 4/5 folds. Alternative recodings give 0.0017 ± 0.0019, improving 3/5 folds. These small, non-uniform gains favor retaining the simpler linear model. Full-batch logistic regression performs similarly to mini-batch training; ridge and the class-weighted SVM score slightly lower. The final CSV generated by `run.py` obtains AIcrowd F1 0.444.

IV. Discussion and conclusion

The strongest measured improvements come from feature coverage and threshold selection. The tested nonlinear extensions offer only small gains, while stronger regularization reduces performance. Individual preprocessing steps were not separately ablated, so their contributions cannot be quantified from these experiments.

The survey is self-reported and cross-sectional: predictors may reflect consequences of disease, and associations do not establish causality or future risk. Feature groups rely on prior judgments. Thresholds use in-sample predictions rather than an inner validation set. Reusing CV folds for model selection can make the reported CV performance optimistic. Nested validation, with threshold selection on an inner held-out subset, would strengthen model comparisons. Within these limits, the results support the simpler groups 1–5 logistic model as the final choice.

REFERENCES

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